

BUSINESS INFORMATION SYSTEMS- PART 1

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Objectives

- Define Information Technology (IT) and Information Systems (IS)
- Describe Information Systems components for business.
- Discuss the impacts IT is having on the business world.
- Categorize business information systems in the context of systems analysis.
- Define communication tools for business data.
- Define networks for business information systems.



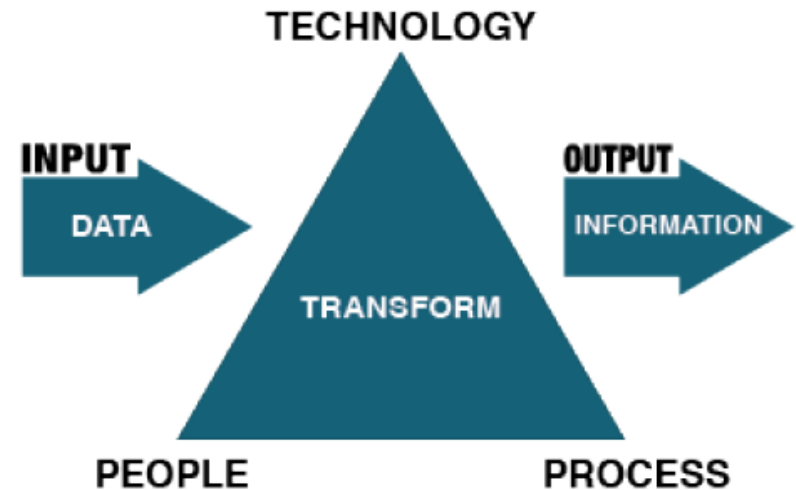
Information Systems

- **Information Systems can be**
 - manual or digital,
 - paper-based or computerized,
 - low tech and low speed, or high tech and high speed.
- **Information Technology (IT) vs Information Systems (IS)**
 - IS contains interrelated components that allow us to take and transform meaningless data into useful information for decision making.
 - IT is a subset, or one component of information systems.



Information Systems Components for Business

- Information Systems can be viewed as having three core components
 - Technology,
 - People,
 - Process that take the data and transform it into information.



Information Systems Components



Information Systems Components-Technology

- Hardware
 - tangible, physical portion of an information system.
 - computers, keyboards, disk drives, and flash drives are all examples of information systems hardware.
- Software
 - the set of instructions that tell the hardware what to do.
 - main categories of software are: Operating Systems and Application software.
- Networking Communication
 - made up of hardware and software



Information Systems Components-People

- The front-line user support staff
- Computer Operator
- Systems analysts
- Developers/Programmers
- Database Administrators
- Managers
- All the way up to the chief information officer (CIO)



Information Systems Components-Process

- Series of steps undertaken to achieve a desired outcome or goal.
- Technology buzzwords such as ‘business process re-engineering,’ ‘business process management,’ and ‘enterprise resource planning’ all have to do with the continued improvement of these business processes and procedures and the integration of technology with them.



Information Systems: The Competitive Power of IT

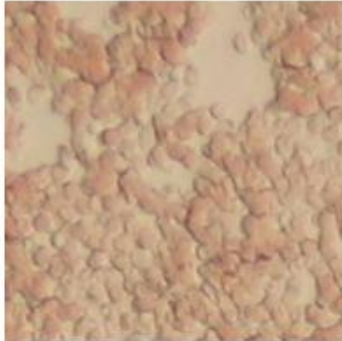
- **Information System (IS)**
 - system that uses IT resources to convert data into information and to collect, process, and transmit that information for use in decision making
- **Data**
 - raw facts and figures that, by themselves, may not have much meaning
- **Information**
 - meaningful, useful interpretation of data



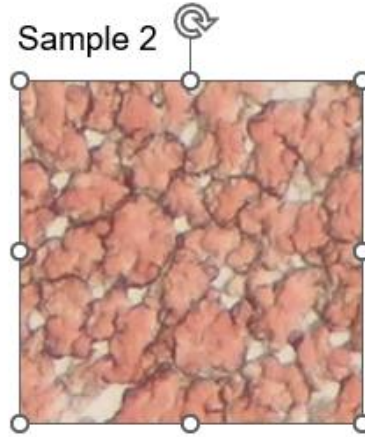
Data vs Information

Blood Sample Images

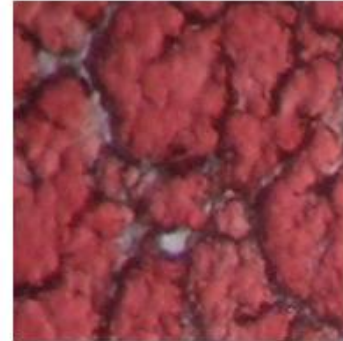
Sample 1



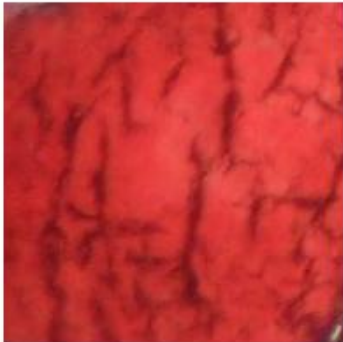
Sample 2



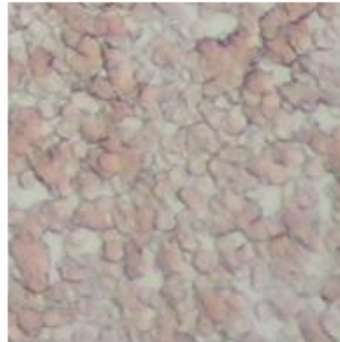
Sample 3



Sample 4



Sample 5

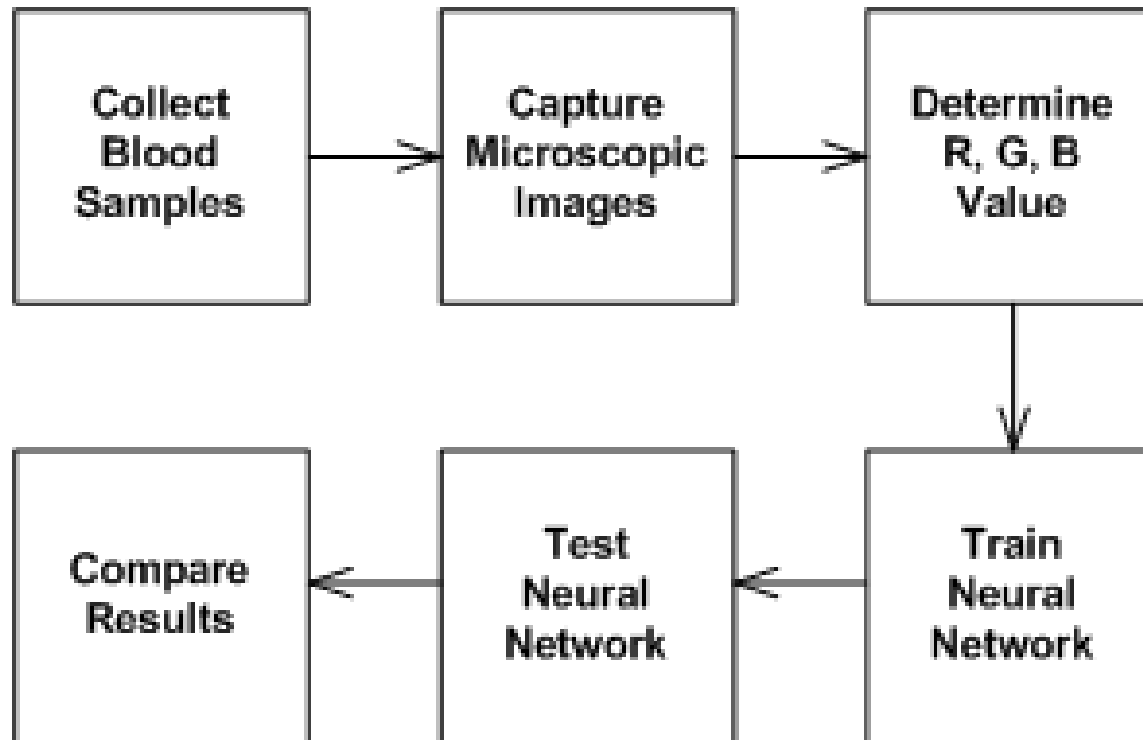


Data vs Information

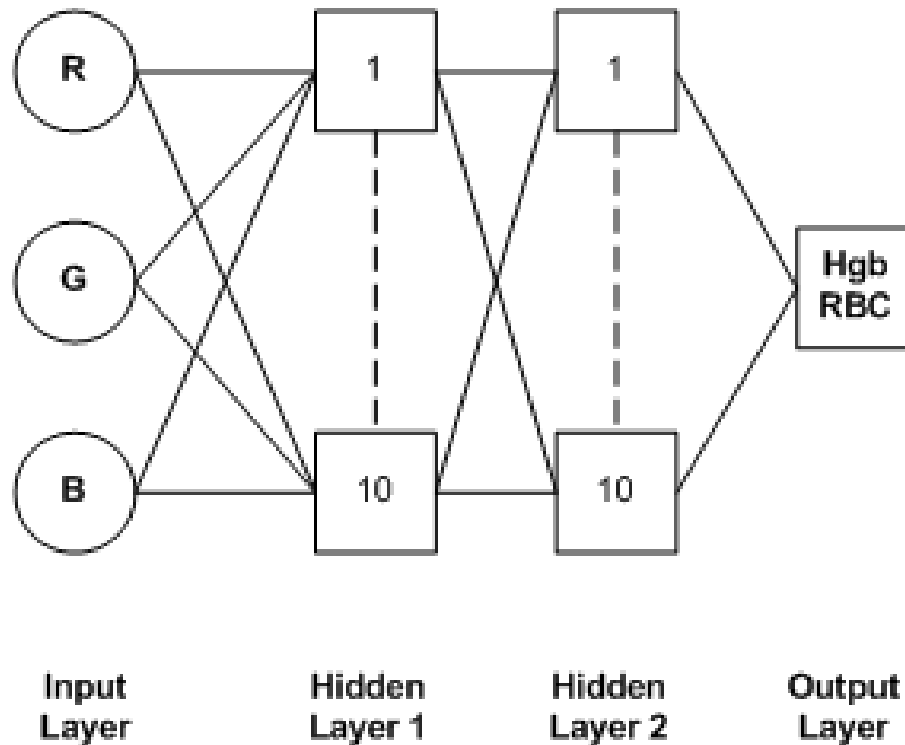
Blood Sample Number	Hgb value		RBC Count		MCH value	
	Lab (g/L)	ANN (g/L)	Lab (10^{12} /L)	ANN (10^{12} /L)	Lab (pg)	ANN (pg)
Sample 1	93.3	98.40	3.17	3.33	29.43	29.54
Sample 2	121	117.94	4.04	3.99	29.95	29.55
Sample 3	152	148.25	4.54	4.44	33.48	33.38
Sample 4	175	168.36	5.42	5.36	32.28	31.41
Sample 5	77.7	73.04	2.56	2.38	30.35	30.68



Extraction of Information from Data



Artificial Neural Network Architecture



Leveraging Information Resources: Data Warehousing and Data Mining

- **Information Systems Managers**
 - managers who are responsible for the systems used for gathering, organizing, and distributing information
- **Data Warehousing**
 - the collection, storage, and retrieval of data in electronic files
- **Data Mining**
 - the application of electronic technologies for searching, sifting, and reorganizing pools of data to uncover useful information



IT Impacts (1 of 3)

- **Information Technology (IT)**
 - various appliances and devices for creating, storing, exchanging, and using information in diverse modes, including visual images, voice, multimedia, and business data
- **E-commerce**
 - use of the Internet and other electronic means for retailing and business-to-business transactions



IT Impacts (2 of 4)

- **Creating Portable Offices**
 - providing remote access to instant information
- **Enabling Better Service**
 - coordinating remote deliveries
- **Creating Leaner, More Efficient Organizations**
 - allowing more work and customer satisfaction to be accomplished with fewer people
- **Enabling Increased Collaboration**
 - using collaboration software and other I T communication devices



IT Impacts (3 of 4)

- **Enabling Global Exchange**
 - fostering collaboration on a worldwide scale
- **Improving Management Processes**
 - using enterprise resource planning (E R P) to change the nature of the management process
- **Providing Flexibility for Customization**
 - creating new manufacturing capabilities that offer customers greater variety (mass customization) and faster delivery cycles



IT Impacts (4 of 4)

- **Providing New Business Opportunities**
 - creating entirely new businesses where none existed before
- **Improving the World and Our Lives**
 - advancing medical and diagnostic techniques
 - Internet



Information Systems for Business

- **Management Information System (MIS)**
 - computer system that supports managers by providing information—reports, schedules, plans, and budgets—that can be used for making decisions
- **Decision Support System (DSS)**
 - interactive system that creates virtual business models for a particular kind of decision and tests them with different data to see how they respond



“Cloud Computing” Model for Business Information Systems

- Model of computing in which computing processing, storage, software, and other services are provided as a shared pool of virtualized resources via the Internet
- Three primary types of services
 - Infrastructure as a service (IaaS)
 - Software as a service (SaaS)
 - Platform as a service (PaaS)
- Public, private, and hybrid clouds



Infrastructure-as-a-service (IaaS)

- Allows organizations to purchase resources like networking and storage on-demand instead of having to buy costly hardware.
- The IaaS provider is responsible for managing and maintaining servers, hard drives, storage, and virtualization tools.
- However, applications, runtime, OS, middleware, and data still need to be managed by the client.



Platform-as-a-service (PaaS)

- Provides developers with a framework and tools to build apps and software that are tailored to the organization's individual needs.
- The IaaS provider is responsible for managing and maintaining servers, hard drives, storage, runtime, OS, middleware and virtualization tools.
- However, applications and data still need to be managed by the client.



Software-as-a-service (SaaS)

- Hosted on remote servers and fully managed, updated, and maintained by a SaaS provider.
- Delivered as a fully functional service and can be accessed remotely via any web browser, allowing clients to work from anywhere.
- Ideal for small companies or startups that don't have the capacity to develop their own software applications.



IaaS vs PaaS vs SaaS

On-site	IaaS	PaaS	SaaS
Applications	Applications	Applications	Applications
Data	Data	Data	Data
Runtime	Runtime	Runtime	Runtime
Middleware	Middleware	Middleware	Middleware
O/S	O/S	O/S	O/S
Virtualization	Virtualization	Virtualization	Virtualization
Servers	Servers	Servers	Servers
Storage	Storage	Storage	Storage
Networking	Networking	Networking	Networking

 You manage  Service provider manages



IaaS vs PaaS vs SaaS

Public Cloud Computing Market Share (Deployment Type), 2019-2022

www.T4.ai



Source: <https://www.t4.ai/industry/cloud-computing-market-share>



Networks for Business Information Systems

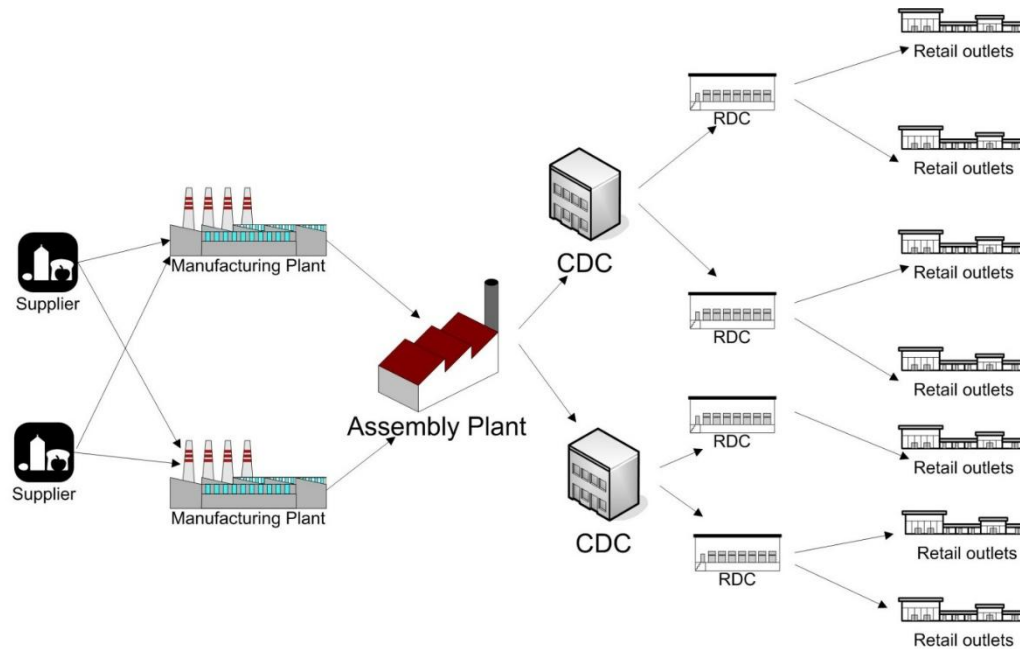


Mobile Internet Access for Business Information Systems

- Emergence of mobile Internet access one of the most significant developments for Internet and Web
- Concerned with the last mile of Internet access to user's home, office, car, smartphone or tablet computer
- Two basic types:
 - Telephone-based (mobile phones, smartphones)
 - Computer network-based (wireless local area network-based)



Wireless Internet Access for Business Information Systems



Wireless Internet Access

- Wireless Access Networks
 - Wireless wide area network (WWAN)
 - ❖ 2G and 3G technology
 - Wireless metropolitan area network (WMAN)
 - ❖ WiMAX technology
 - Wireless local area network (WLAN)
 - ❖ Wi-Fi technology
 - Wireless personal area network (WPAN)
 - ❖ UWB technology

Wireless Internet Access for Business Information Systems

- Connects user to global telephone system (land, satellite, and microwave)
- 4G: true broadband access, speeds up to 100 Mbps
- 5G: very high-bandwidth broadband; speeds reaching 10 Gbps or more
 - Uses new part of wireless spectrum and tens of thousands of small-cell and distributed antenna systems
 - Expected to enable many innovative products. services

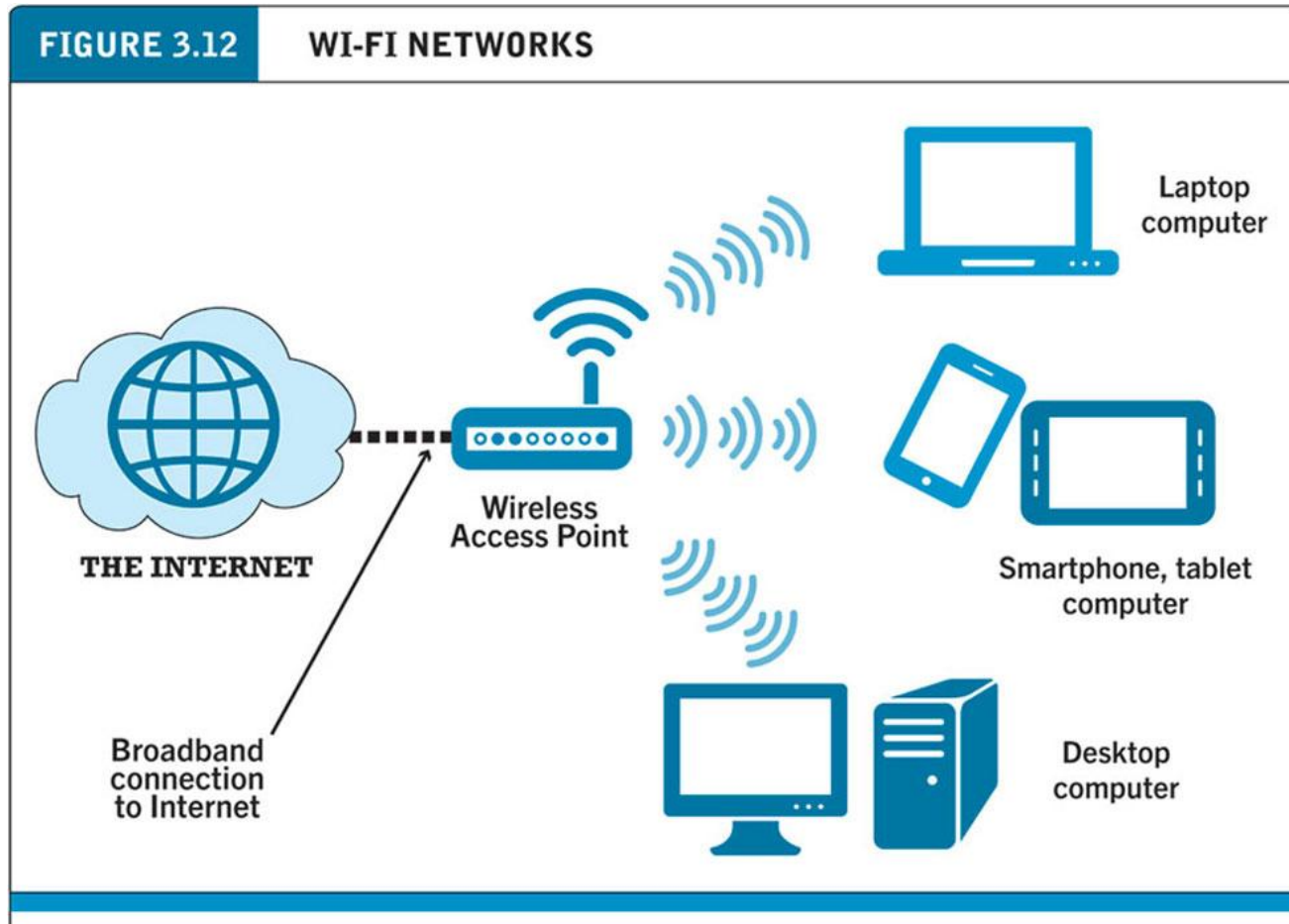


Wireless Local Area Network (WLAN) - Based Internet Access for Business Information Systems

- Wi-Fi: various 802.11 standards, also known as Wi-Fi 4, Wi-Fi 5, and Wi-Fi 6
 - High-speed, fixed broadband wireless LAN (WLAN)
 - Wireless access points (“hot spots”)
 - Limited range but inexpensive
- WiMax: 802.16 standard; used for medium-range, broadband wireless metropolitan area networks
- Bluetooth: technology standard for short-range wireless communication within a 30-foot distance



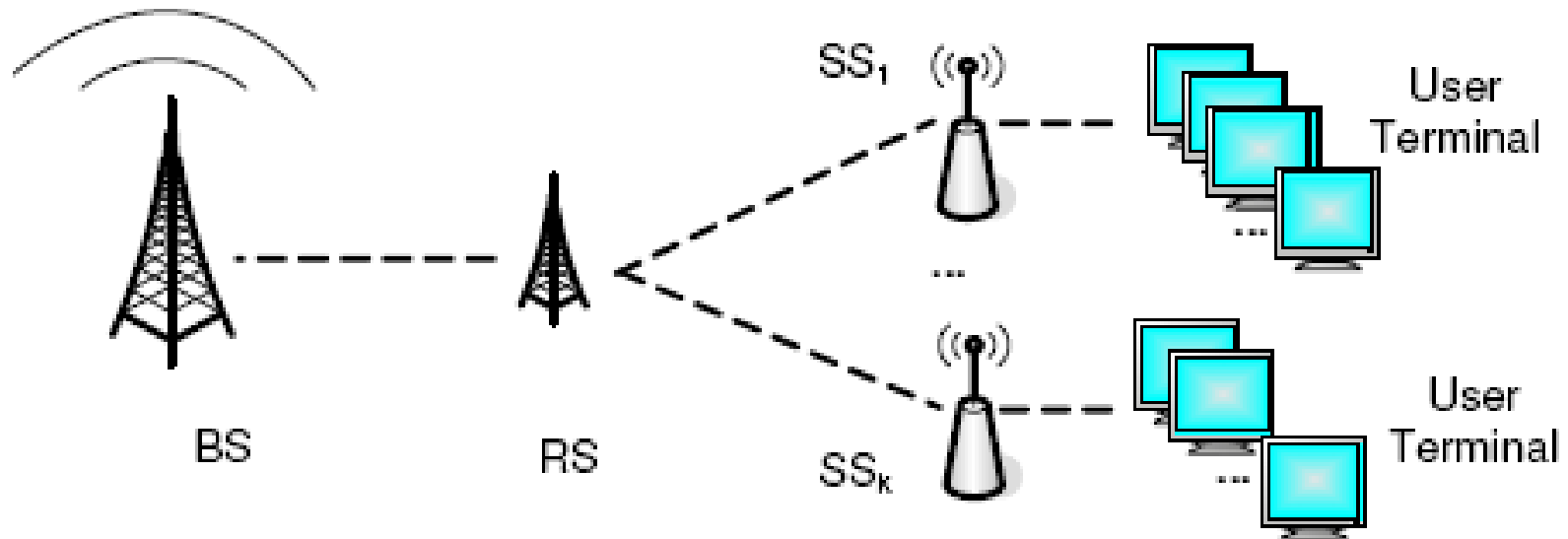
Wi-Fi Networks



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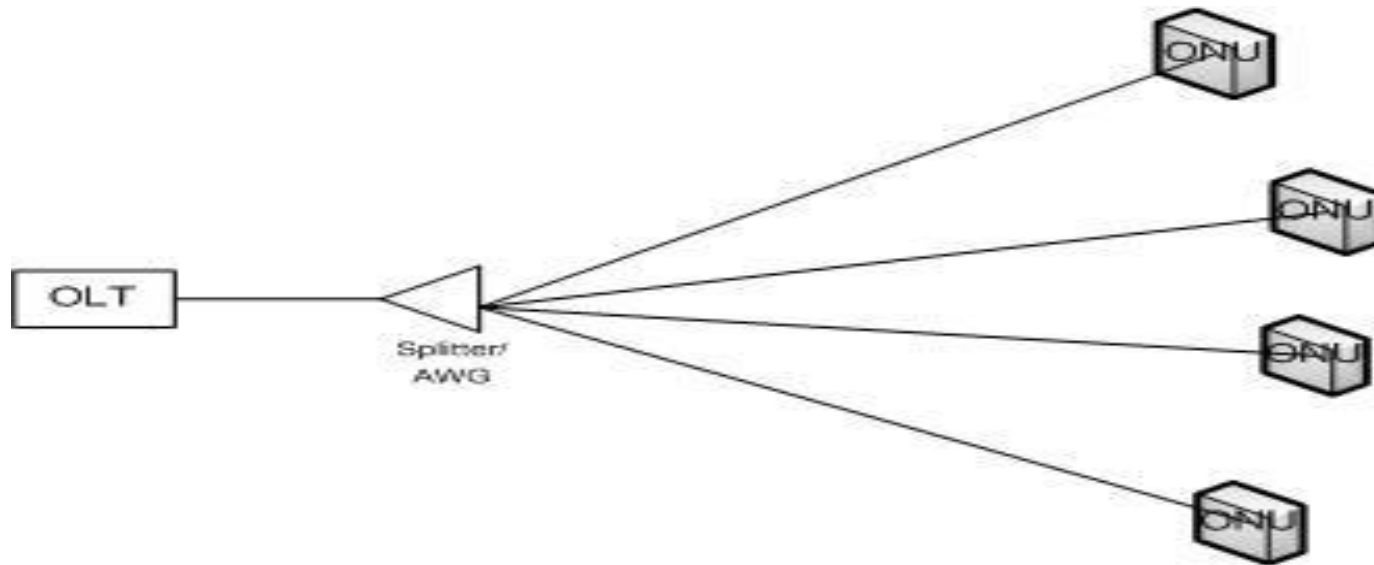
WiMAX Networks



- WiMAX BS
- Subscriber station (SS) or Customer Premise Equipment (CPE)
- Relay Station (RS)



Optical Access Networks



OLT- Optical Line Terminal

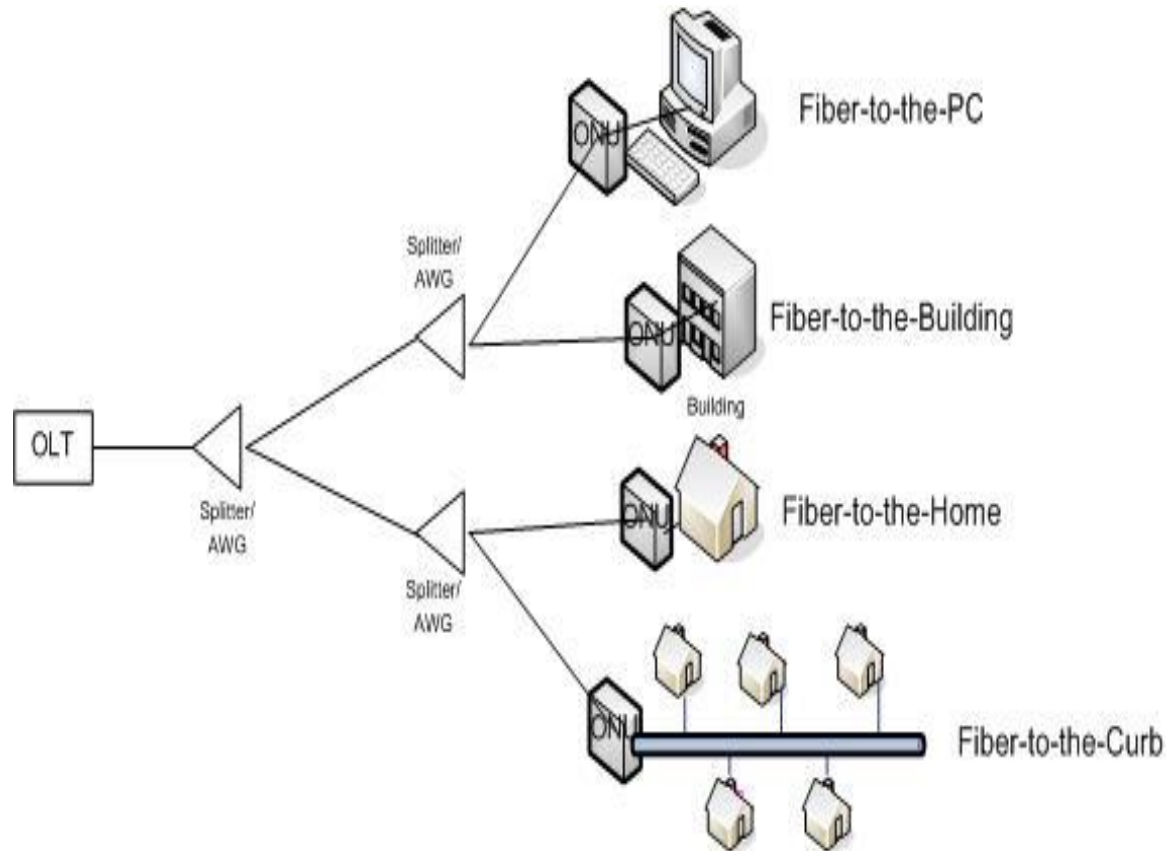
AWG- Arrayed Waveguide Grating

ONU- Optical Network Unit

Splitter- Splits the incoming optical signal into multiple outgoing links



Optical Access Networks



OLT- Optical Line Terminal

AWG- Arrayed Waveguide Grating

ONU- Optical Network Unit

Splitter- Splits the incoming optical signal into multiple outgoing links



Hybrid Optical-Wireless Access Networks

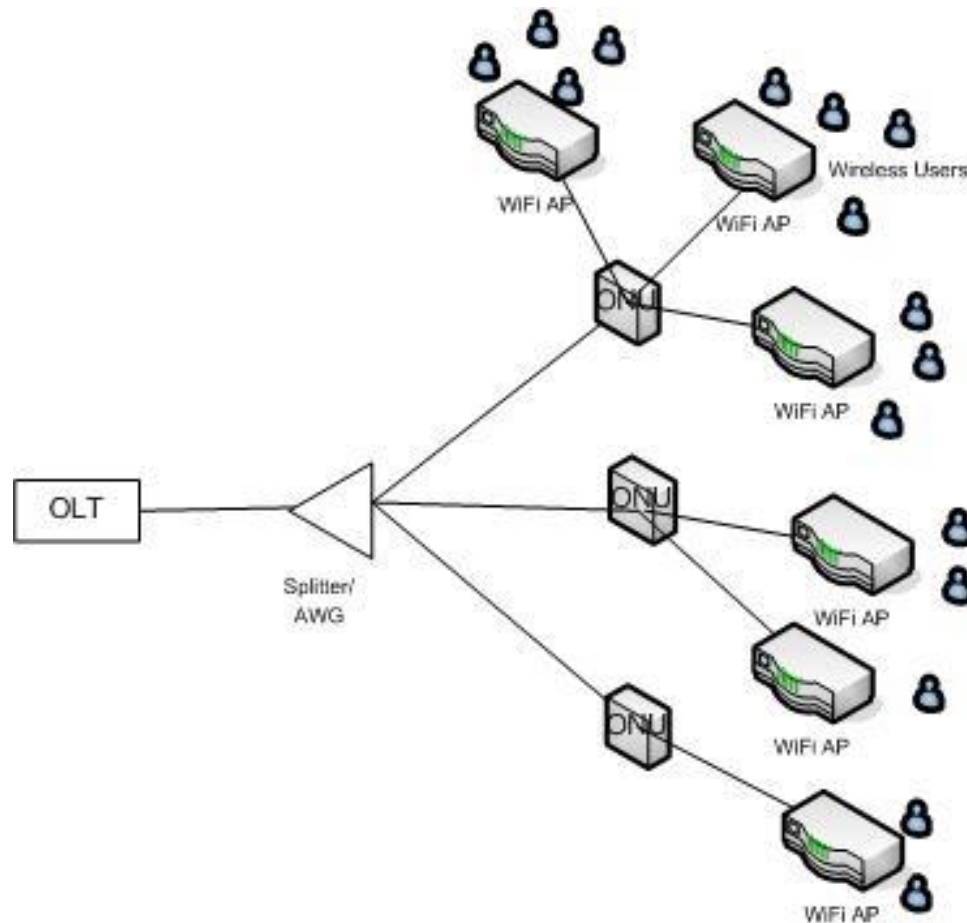
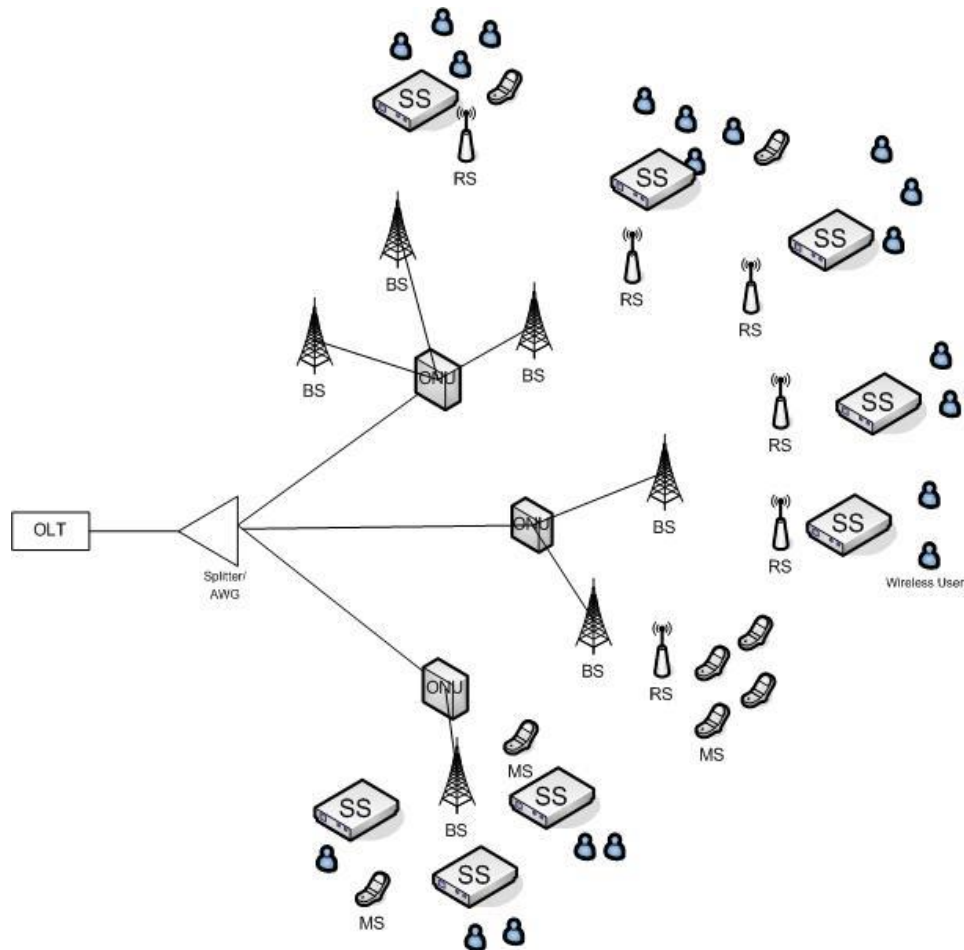


Fig. 2. PON-WiFi Architecture



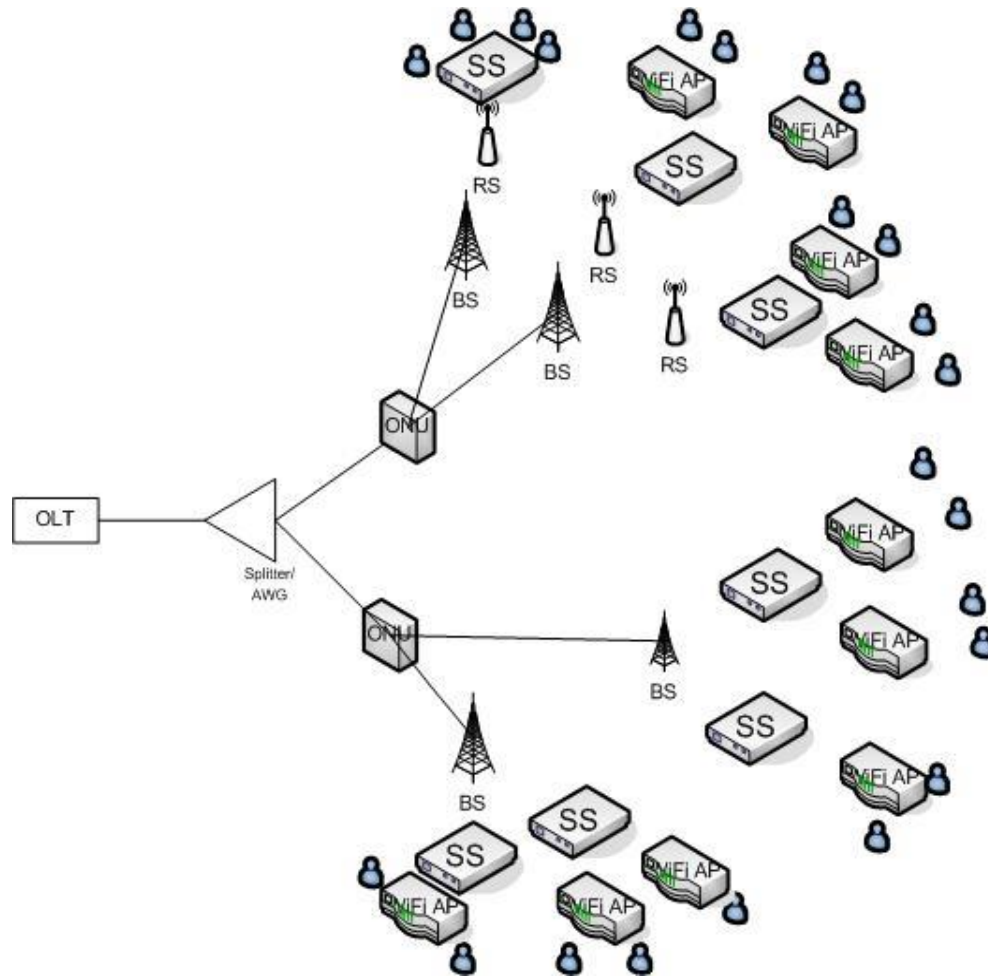
Hybrid Optical-Wireless Access Networks



PON-WiMAX Architecture



Hybrid Optical-Wireless Access Networks



PON-WiMAX-WiFi Architecture



Hybrid Optical-Wireless Access Networks

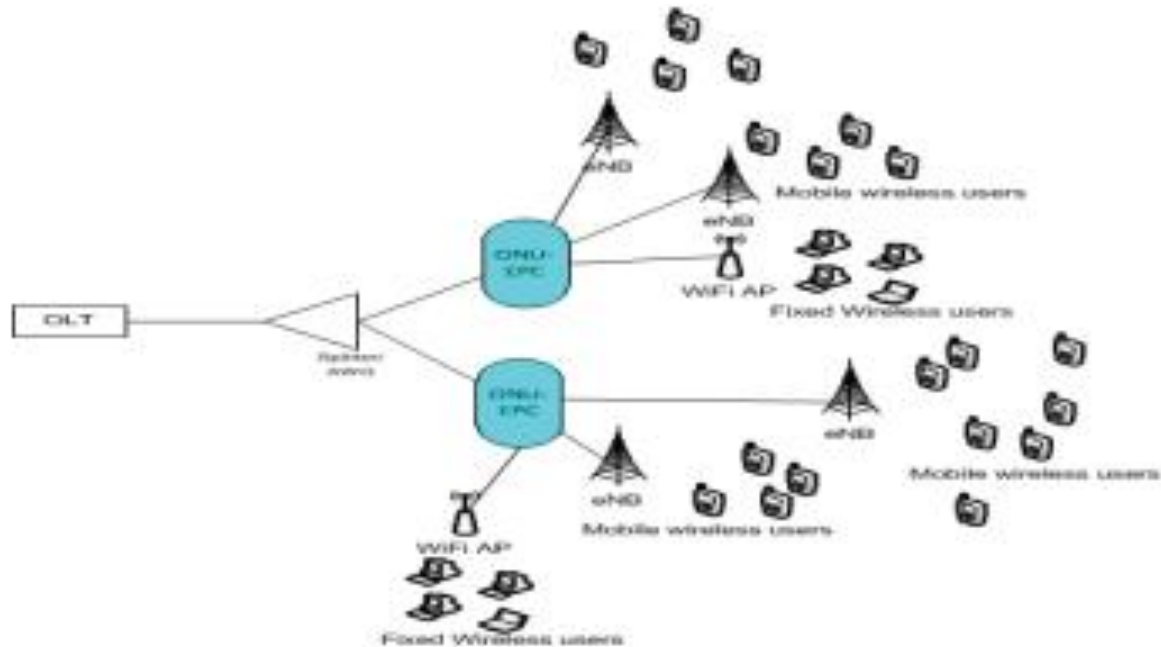
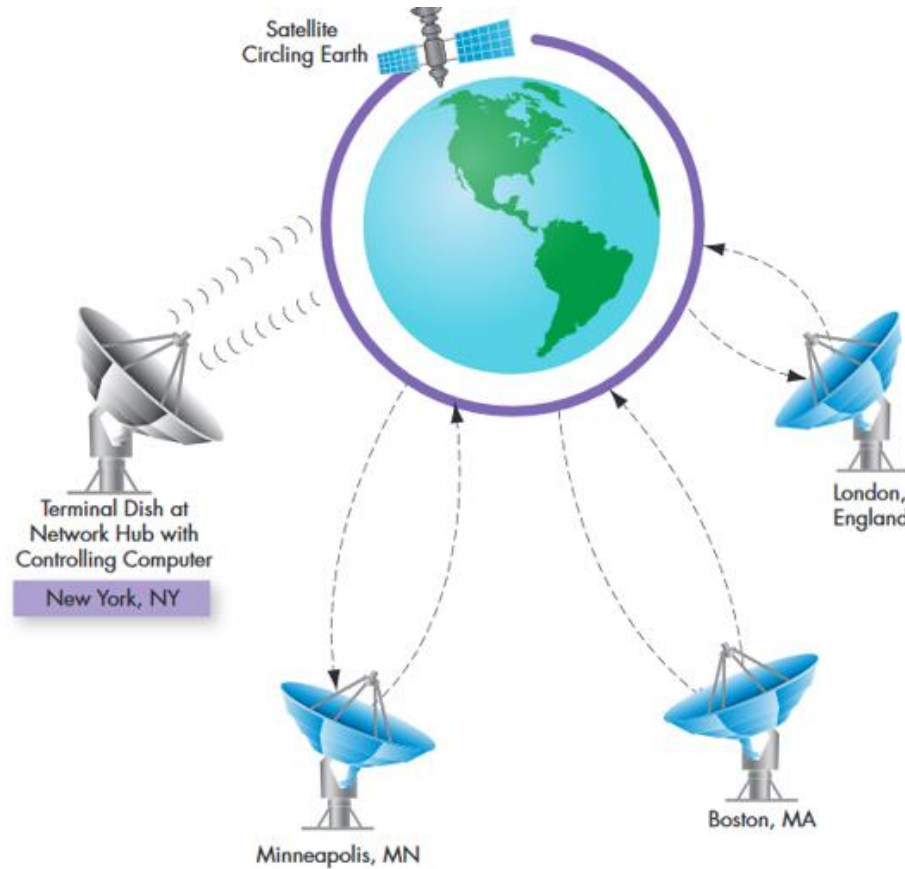


Fig. 5. PON-LTE-WiFi Architecture



A VSAT Satellite Communication Network



The Internet of Things (IoT) for Business Information Systems

- Objects connected via sensors/RFID to the Internet
- Being enabled by:
 - Availability of low-cost sensors
 - Drop in price of data storage
 - Development of big data analytics software
 - Implementation of IPv6
- Powering development of “smart” connected things
- Interoperability issues and standards
- Security and privacy concerns



Web Servers and Web Clients for Business Information Systems

- Web server software
 - Enables a computer to deliver HTML web pages to clients on a network that request this service by sending an HTTP request
 - Basic capabilities: HTTP, security services (TLS), file transfer, search engine, data capture, e-mail, site management tools
- Web server
 - May refer to either web server software or physical server
 - Specialized server software: Database server, ad server, mail server, media server
- Web client
 - Any computing device attached to the Internet capable of making HTTP requests and displaying HTML pages



Virtual Reality, Augmented Reality, and the Metaverse for Business Information Systems (1 of 2)

- Virtual reality (VR)
 - Immersing users within virtual world
 - Typically uses head-mounted display (HMD)
 - Oculus Rift, Vive, PlayStation VR
- Augmented reality (AR)
 - Overlaying virtual objects over the real world, via mobile devices or HMDs
 - Pokémon GO
- Mixed reality (AR)
 - Enhanced version of AR



Virtual Reality, Augmented Reality, and the Metaverse for Business Information Systems (2 of 2)

- Metaverse
 - Immersive, visual 3-D virtual reality in which users can connect, socialize, and collaborate
 - Facebook has rebranded as Meta
 - Tremendous hype
 - Currently being used for gaming and advertising
 - Many companies are working on metaverse-related projects



Business Communications

- **Electronic Conferencing**
 - IT that allows groups of people to communicate simultaneously from various locations via e-mail, phone, or video
- **VSAT Satellite Communications**
 - network of geographically dispersed transmitter-receivers (transceivers) that send signals to and receive signals from a satellite, exchanging voice, video, and data transmissions



Communication Tools for Business Data

- E-mail
 - Most used application of the Internet
- Messaging Applications
 - Instant messaging
- Online message boards
- Internet telephony
 - VOIP
- Videoconferencing, video chatting, telepresence



References

- [IaaS vs. PaaS vs. SaaS - Differences, Examples and Diagram | LeanIX](#)

